GRADE 11 · ALGEBRA 2 · MATHEMATICAL MODELING INVESTIGATION

Systems

Under Pressure

Investigation Notebook

DRIVING QUESTION

How do we know when a system is actually in danger?

GROUP NAME

MEMBERS

INVESTIGATION TOPIC

CATEGORY

ESSENTIAL QUESTION

NOTEBOOK STRUCTURE

HOW TO USE THIS NOTEBOOK

This notebook is your group's shared investigation journal. Individual handwritten extensions are completed separately on paper and shared during the vlogs.

The group records 3 informal vlog conversations, one for each unit: Sequences & Series, Exponentials & Logarithms, and Polynomials.

Each student shows and explains one handwritten individual extension during the vlog. There is no daily vlog after every class and no required 150-word reflection per vlog.

CLAIM -> EVIDENCE -> TEST -> REVISION

Making Sense of Our Topic

Use this page to choose a focused investigation question your group can study with data.

OPTIONAL FLINT SUPPORT

THINK FIRST -> FLINT -> VERIFY -> DECIDE

Use Flint only after your group thinks first. Ask: Who is affected, and what evidence could show pressure in this system?

Ask Flint for perspectives you may have missed, then verify any factual claims with sources.

WHO IS AFFECTED - AND WHAT DO THEY EXPERIENCE?

WHO HOLDS POWER OVER THIS SYSTEM?

WHAT WE THINK WE KNOW

WHAT CONFUSES OR SURPRISES US

WHAT MAKES THIS URGENT?

OUR GROUP INVESTIGATION QUESTION - DRAFT

Empathy + Multiple Perspectives

Use this page to decide whose experience should shape your variables and thresholds.

OPTIONAL FLINT SUPPORT

THINK FIRST -> FLINT -> VERIFY -> DECIDE

Use Flint only after your group lists perspectives first. Ask it to challenge your list, not replace it.

Ask Flint what variables different stakeholders might care about, then decide which ones are measurable.

STAKEHOLDER 1 - WHO THEY ARE + WHAT THEY EXPERIENCE

STAKEHOLDER 2 - WHO THEY ARE + WHAT THEY EXPERIENCE

WHAT ALL PERSPECTIVES SHARE

WHERE PERSPECTIVES DISAGREE

VARIABLES WORTH MEASURING - NAMED BY STAKEHOLDERS

Variables + Hidden Systems

Use this page to map variables before deciding what causes what.

OPTIONAL FLINT SUPPORT

THINK FIRST -> FLINT -> VERIFY -> DECIDE

Use Flint only after your group draws an initial systems map. Ask what hidden variables or delays may be missing.

Ask Flint for possible confounding variables, then mark which ones you can actually investigate.

CONNECTED SYSTEMS MAP - DRAW ARROWS, PRESSURES, CONNECTIONS

VARIABLES WE CAN MEASURE

VARIABLES WE CANNOT EASILY MEASURE

REQUIRED SENTENCE FRAME: These variables appear related because ___, but this does not prove causation because ___.

GROUP ______________________ DATE ______________________

Sources + Data Tracker

Use this page to record sources that can support graphs, calculations, and claims.

Source / URL	Variable or Data	Timeframe	Credibility + Limitations	Decision

SOURCE WE TRUSTED - WHY

SOURCE WE QUESTIONED OR REJECTED - WHY

QUESTIONS MY SOURCES CANNOT ANSWER

Graph Observations

Use this page to decide what the graph shows before you build a model.

PRIMARY GRAPH - PASTE, SKETCH, OR ANNOTATE

TRENDS I NOTICE

THRESHOLDS I CAN IDENTIFY

TURNING POINTS

ANOMALIES OR SURPRISES

GROUP

DATE

Local vs Global Behavior

Use this page to choose the interval that tells the most honest mathematical story.

SHORT INTERVAL - LOCAL BEHAVIOR

LONG INTERVAL - GLOBAL BEHAVIOR

WHAT CHANGES WHEN I ZOOM OUT?

WHAT PATTERNS ONLY APPEAR LONGER-TERM?

WHICH INTERVAL TELLS THE MOST HONEST STORY?

Sequences & Series Modeling

Use this page to model repeated change and decide whether arithmetic or geometric reasoning fits.

n	value	difference	ratio	Stable?	Notes / context

PATTERN TYPE + JUSTIFICATION

RECURSIVE MODEL

$a_n = r \cdot a_{n-1} + d$

EXPLICIT MODEL

$a_n = a_1 r^{n-1}$

SERIES / PARTIAL SUM

$S_n = a_1(1 - r^n)/(1 - r)$

PREDICTION: VALUE AT n = ___

INTERPRETATION IN CONTEXT

WHERE THE MODEL BREAKS OR BECOMES UNREALISTIC

Exponentials + Thresholds

Use this page to test a ratio, build an exponential model, and solve for a threshold.

x	y value	ratio	Stable?	Notes / context

RATIO TEST: IS b STABLE ENOUGH?

MODEL EQUATION

$f(x) = a \cdot b^x$

INTERPRET a AND b IN CONTEXT

THRESHOLD SOLVE

$\log_b(\text{threshold}/a) = x$

PREDICTION + UNITS

WHERE EXPONENTIAL REASONING FAILS

Polynomial Behavior

Use this page to model local graph shape: zeros, turns, thresholds, and multiplicity.

GRAPH SKETCH - LOCAL INTERVAL

KEY FEATURES: zeros, thresholds, turns, max/min

ZEROS / THRESHOLDS

TURNING POINT

MULTIPLICITY MEANING

FACTORED FORM

$$P(t) = a(t - r_1)(t - r_2)^2(t - r_3)\ldots$$

WHAT DOES THIS MODEL REVEAL THAT A LINEAR OR EXPONENTIAL MODEL MISSES?

Model Comparison

Use this page to compare what each model predicts, reveals, hides, and assumes.

Criterion	Linear	Exponential	Polynomial	Sequence / Series
Equation/model used				
What it predicts				
What it reveals				
What it hides				
Assumption needed				
Where it fails				

MOST DEFENSIBLE MODEL - AND WHY THE OTHERS ARE LESS APPROPRIATE

GROUP ______ DATE ______

Confounding Variables

Use this page to test whether another variable could explain your pattern.

CONFOUNDING VARIABLE 1 - NAME + EFFECT

CONFOUNDING VARIABLE 2 - NAME + EFFECT

CONFOUNDING VARIABLE 3 - NAME + EFFECT

CONFOUNDING VARIABLE 4 - NAME + EFFECT

REQUIRED SENTENCE FRAME: These variables appear related because ___, but this does not prove causation because ___.

GROUP

DATE

Optional Flint Critique Log

Use this page only when Flint helps you test, critique, or revise mathematical reasoning.

PROTOCOL: THINK FIRST -> FLINT -> VERIFY -> DECIDE

ENTRY 1: prompt, suggestion, accepted/rejected, verification

ENTRY 2: prompt, suggestion, accepted/rejected, verification

ENTRY 3: prompt, suggestion, accepted/rejected, verification

GROUP

DATE

Revision Trail

Use this page to show exactly how evidence changed your group's thinking.

ORIGINAL CLAIM

EVIDENCE THAT CONTRADICTED IT

REVISED INTERPRETATION - WHAT CHANGED, WHY, AND WHAT EVIDENCE FORCED IT

REMAINING UNCERTAINTY

GROUP

DATE

Final Claim

Use this page to connect your final claim to specific evidence, calculations, and rejected models.

WHICH GRAPH SUPPORTS YOUR CLAIM?

WHICH CALCULATION SUPPORTS YOUR CLAIM?

WHICH MODEL DID YOU REJECT - AND WHY?

WHAT EVIDENCE WOULD CHANGE YOUR CONCLUSION?

OUR BIGGEST LIMITATION

REMAINING UNCERTAINTY

OUR FINAL MATHEMATICAL CLAIM

GROUP ______ DATE ______

Presentation Prep

Use this page to plan a mathematical argument supported by graphs, calculations, and model choices.

OUR CENTRAL CLAIM - ONE SENTENCE

WHICH GRAPH SUPPORTS THE CLAIM?

WHICH CALCULATION SUPPORTS THE CLAIM?

WHICH MODEL DID WE REJECT - AND WHY?

WHAT EVIDENCE WOULD CHANGE OUR CONCLUSION?

OUR BIGGEST LIMITATION + FOLLOW-UP QUESTION WE EXPECT

GROUP ______________________ DATE ______________